Video-Behavior Alignment Report

Animal: 4_Milka | Session: R1 | Date: 2023-08-15

Run Settings

Parameter	Value
Script	video_alignment_run_ready.py
--frame-step	10 (180,745 frames sampled)
--sweep	15-config grid search
Runtime	3-4 hours

Hyperparameter Search Results

Grid: min_conf ∈ {0.2, 0.3, 0.35, 0.4, 0.5} × pc1_weight ∈ {0.3, 0.45, 0.6} (15 configurations, 128 valid trial pairs each).

Rationale: Greater mean confidence decreases the probability of false positives while increasing the probability of false negatives. It is very arbitrary and dependent on the specific emsemble and intractable ways that the feature of the kernel varies and is accounted for in the pc1 and temple matching, so empirically test and later use the value that produced the best results for R1. The Pc1's accuracy relative to the emsemble is also quite unpredictable, so we also try out different values.

Best configuration: min_conf = 0.3, pc1_weight = 0.6 → MAE = 3.09 s, RMSE = 3.68 s

min_conf	pc1_w	tpl_w	n_valid	n_train	n_test	stretch	MAE (s)	RMSE (s)
0.3	**0.6**	**0.4**	**128**	**102**	**26**	**1.0023**	**1.064 ★**	**151.08**
0.35	0.3	0.7	128	102	26	1.0015	2.686	151.18
0.2	0.45	0.55	128	102	26	1.0013	3.672	151.12
0.4	0.6	0.4	128	102	26	1.0019	1.593	151.14
0.3	0.45	0.55	128	102	26	1.0030	-0.236	150.95
0.2	0.3	0.7	128	102	26	1.0021	1.150	151.07
0.5	0.3	0.7	128	102	26	1.0020	1.436	151.06
0.2	0.6	0.4	128	102	26	1.0023	0.579	151.03
0.5	0.6	0.4	128	102	26	1.0014	2.386	151.15
0.35	0.6	0.4	128	102	26	1.0013	2.524	151.23
0.35	0.45	0.55	128	102	26	1.0012	1.868	151.30
0.3	0.3	0.7	128	102	26	1.0021	0.997	150.99

Metric	Value

Warp stretch	1.002
Warp offset	1.15 s
Clock drift (total)	~4.8 s over 2300 s
Initial bi[0]	1.099 (hit bound)
Trials aligned	**138 / 138 (100%)**

Confidence Score Distribution

Confidence scores reflect ensemble detection quality (PC1 peak + template match + isolation). Since the run with the highest PC1 performed the best, in future runs we will use it as the strict metric.

Metric	Value
Min confidence	0.51 (trial 129)
Max confidence	1.00 (trials 1, 8, 26, 41, 113, 124)
Median confidence	~0.80

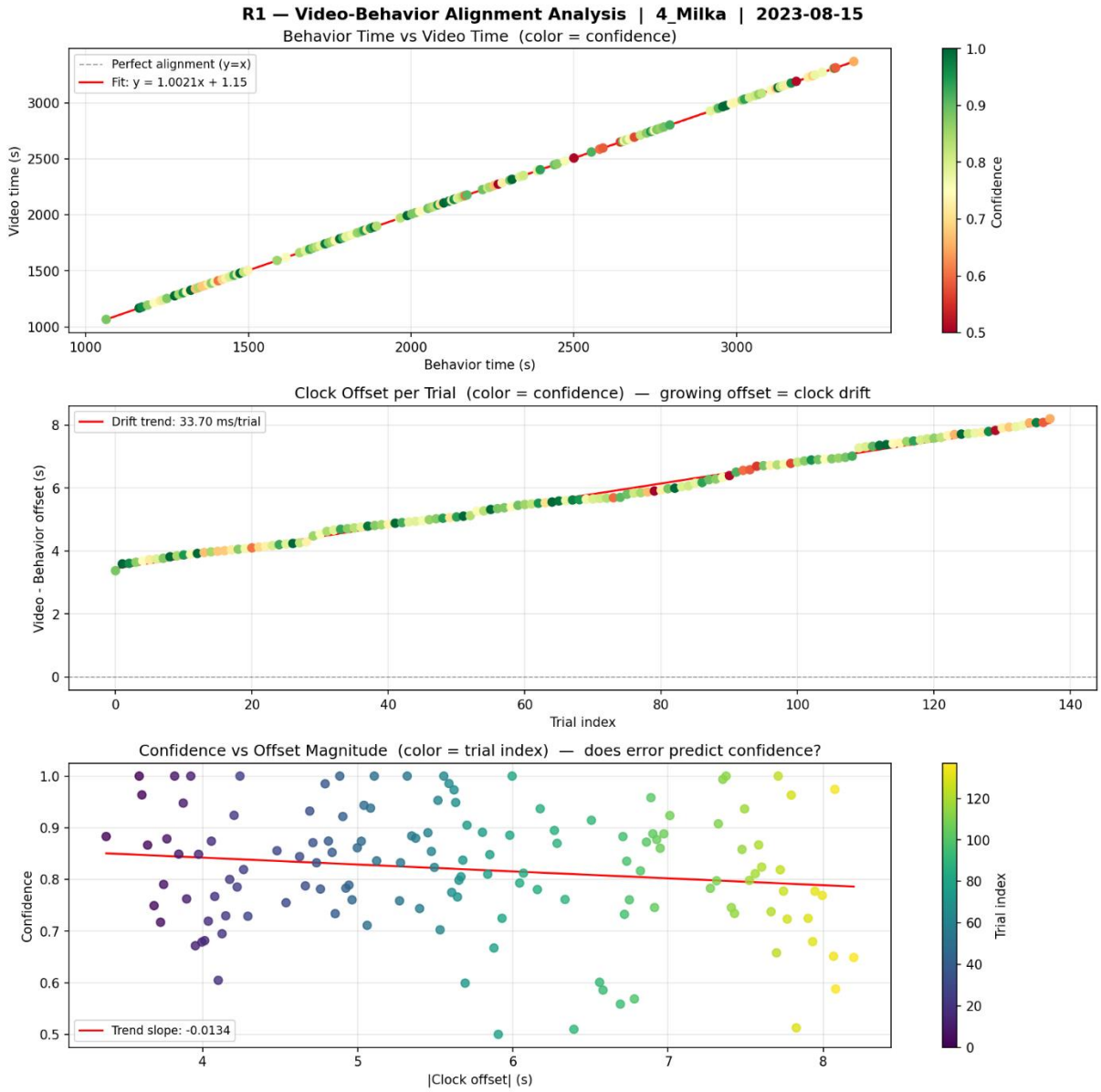
Per-Trial Alignment Table (sampled)

Video time = behavior time + clock offset + drift. Drift increases slowly over the session (~0.2%/s).

Trial	Bhv time (s)	Video time (s)	Offset (s)	Confidence
0	1062.37	1065.75	+3.38	0.883
1	1164.07	1167.67	+3.60	**1.000**
2	1172.31	1175.92	+3.61	**0.964**
10	1299.34	1303.22	+3.88	**0.948**
20	1406.99	1411.10	+4.11	0.605
30	1615.36	1619.89	+4.53	0.755
41	1780.53	1785.41	+4.88	**1.000**
50	1905.62	1910.99	+5.37	0.831
60	2055.02	2060.96	+5.94	0.718
70	2192.27	2198.60	+6.33	0.857
80	2363.27	2369.87	+6.60	0.716
90	2527.53	2534.00	+6.47	0.755
100	2704.72	2711.54	+6.82	0.816
110	2938.73	2946.04	+7.31	0.797
120	3068.21	3075.79	+7.58	0.867
130	3220.20	3228.10	+7.90	0.725

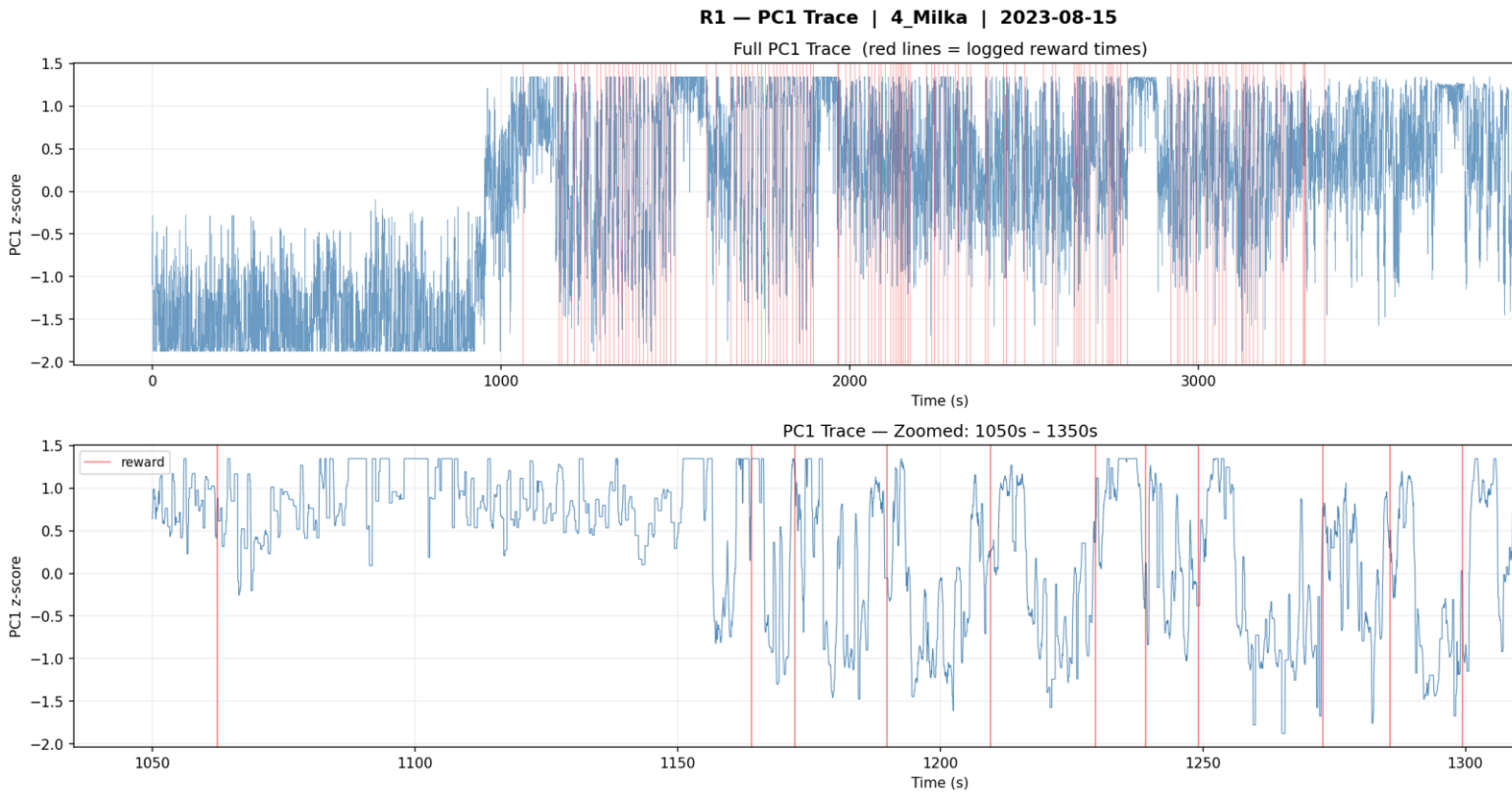
137	3360.44	3368.64	+8.20	0.649

Visualization



Plot 1,2,3: Obtained linear warp compared to the behaviour log times, error (offset) between both as function of time (trials are ordered) and comparison of confidence and the size of the error, respectively.

The last plot shows that there is a slight agreement between the correlation/synchrony of the different features and the accuracy of the prediction.



Plots 4 and 5 show the value of PC1 against the reward times. We notice the drops in the PCA consistent with reward times as expected and we can clearly distinguish the non active portion of the session before the 100s mark.

Output Files

File	Contents
sync_Camera1...json	Canonical sync: warp params + video_time per trial + confidence scores
hp_sweep_Camera1...csv	Full 15-config hyperparameter sweep and results
Camera1..._diagnostics.npz	pc1_z trace in time

Known Issues & Next Steps

Issue	Impact	Fix
Offset bound hit (±5s)	Initial warp suboptimal, high warp_rmse=151	Widen to ±100s before R2-R4
tp/fp/fn = 0	Errors too large made the confusion matrix trivial	Improve MAE

Clock drift grows	~4.8s total drift over session	handled by warp model

R2 run: min_conf=0.3, pc1_w=0.6

R3, R4: Use only pc1 instead of emsemble voting score, and then use only falling edge detection

Pending fix: Widen offset bounds to ±100s in fit_time_warp()

Final goal: Apply pipeline to ephys↔behavior sync (primary scientific problem)